

EF5342 Financial Systems, Markets and Instruments

Week 8 — Derivatives

Prof. Weikai Li

City University of Hong Kong

Semester B 2026–2027

What Are Derivatives, and Why Trade Them?

Derivatives are securities whose value is derived from an underlying asset or financial instrument — a stock, T-bill/bond, foreign currency, commodity, or even another derivative. They generally involve an agreement between two parties to exchange a quantity of an asset or cash flow at a predetermined price at a specified future date. Main types: **forwards, futures, options, swaps**.

Speculate Profit from expected price changes in the underlying; no position in the underlying asset

Hedge Minimize or manage risk; hold a position in the underlying with the goal of offsetting its risk

Arbitrage Exploit price discrepancies between the underlying and its derivatives (e.g., covered interest rate parity arbitrage); simultaneous positions in both

A derivative contract is a **zero-sum game**: what you gain, your counterparty loses.

Forwards vs. Futures

Spot contract Agreement to transact involving the immediate exchange of assets and cash

Forward contract Non-standardized agreement to buy or sell an asset in the future, with terms set today: customized, not traded in a secondary market (participants must perform), with **counterparty risk**

Futures contract Standardized, exchange-traded version of a forward

Futures differ from forwards:

- **Liquid** — most traders close out before delivery, so delivery may never take place
- **Marked to market daily** — gains and losses realized each day as futures prices change
- **Clearing house** stands behind all contracts — minimal counterparty default risk
- **Margin requirements** — a deposit ensuring the terms of the contract will be met

Post-crisis reform: since the 2009 G20 Pittsburgh commitment, standardized OTC derivatives (e.g., interest-rate and credit-default swaps) must also clear through central counterparties — implemented via Dodd-Frank Title VII (2010) in the US and EMIR (2012) in Europe.

Futures Contracts: Mechanics

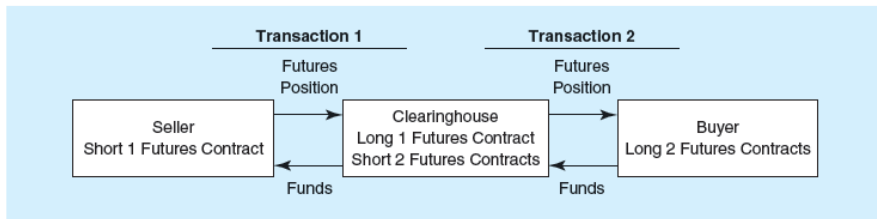
Long position Purchase of a futures contract

Short position Sale of a futures contract

Open interest Total number of futures contracts outstanding at the beginning of the day

Clearing house The unit that oversees trading on the exchange and guarantees all trades — buyer to every clearing-member seller, seller to every buyer

Figure 10-3 Clearinghouse Function in Futures Markets



Underlyings and Settlement

The underlying asset can be T-notes/bonds, currency, commodity, stock index, single stock, or bitcoin (spot bitcoin ETFs and listed options on them are now mainstream exchange-traded products). Settlement is of two kinds:

Physical delivery The actual underlying asset is delivered on the specified delivery date

Cash settlement Counterparties net out the cash difference in the value of their positions; no asset is delivered at expiration

Foreign Currencies	Agricultural	Metals and Energy	Interest Rate Futures	Equity Indexes
British pound	Corn	Copper	Eurodollar	S&P 500 index
Canadian dollar	Oats	Aluminum	Euroyen	Dow Jones Industrials
Japanese yen	Soybeans	Gold	Euro-denominated bond	S&P Midcap 400
Euro	Soybean meal	Platinum	Euroswiss	NASDAQ 100
Swiss franc	Soybean oil	Palladium	Sterling	NYSE index
Australian dollar	Wheat	Silver	British government bond	Russell 2000 index
Mexican peso	Barley	Crude oil	German government bond	Nikkei 225 (Japanese)
Brazilian real	Flaxseed	Heating oil	Italian government bond	FTSE index (British)
	Palm oil	Gas oil	Canadian government bond	CAC-40 (French)
	Rye	Natural gas	Treasury bonds	DAX-30 (German)
	Cattle	Gasoline	Treasury notes	All ordinary (Australian)
	Hogs	Propane	Treasury bills	Toronto 35 (Canadian)
	Pork bellies	Kerosene	LIBOR	Dow Jones Euro STOXX 50
	Cocoa	Fuel oil	EURIBOR	Industry indexes:
	Coffee	Iron ore	Interest rate swaps	• Banking
	Cotton	Electricity	Federal funds rate	• Telecom
	Milk	Weather	Bankers' acceptance	• Utilities
	Orange juice			• Health care
	Sugar			• Technology
	Lumber			
	Rice			

Example Contracts from CME

Physically settled: Light Sweet Crude Oil (WTI) futures — cmegroup.com

Cash-settled: E-mini Crude Oil futures — cmegroup.com

Energy Products
View an Energy Product
Select product

Light Sweet Crude Oil
Quotex | Contract Specifications | Performance Bonds / Margins | Product Calendar | Learn More

Education | **News** | **Description**

Light Sweet Crude Oil (Physical) futures are an outright crude oil contract between a buyer and seller. The contracts also serve as a key international pricing benchmark, and:

- Offer excellent liquidity and price transparency
- Provide the world's most liquid forum for crude oil trading
- Are the world's largest-volume futures contract on a physical commodity
- Serve the diverse needs of the physical market

Things to know:

- Unit of trading is 1,000 barrels
- Delivery point is Cushing, Oklahoma, which is also accessible to the international spot markets via pipelines
- Delivery provided for several grades of domestic and internationally traded foreign crudes
- Six types of options: American style, calendar spread, crack spread, average price, European style and daily

About Light Sweet Crude Oil
Crude oil is the world's most actively traded commodity. Light, sweet crudes are preferred by refiners because of their low sulfur content and relatively high yields of high-value products such as gasoline, diesel fuel, heating oil, and jet fuel.

NYMEX, a member exchange of CME Group, also lists for trading electronically a financially settled futures contract for Dubai crude oil, a futures contract on the differential between the light, sweet crude oil futures contract and Canadian Base River crude oil, and futures contracts on the differentials of the light, sweet crude oil futures contract and four domestic grades of crude oil: Light Louisiana Sweet,

Most Requested

- Energy Market Commentary
- Energy Update - Archive
- Excessive Speculation and Position Limits in Energy Derivatives Markets
- Free Schedule
- Free Real-time Quotes
- Learn more about CME Clearport
- Login to CME ClearPort
- Margins / Performance Bonds
- Monthly Energy Update
- NYMEX Daily Reports
- NYMEX Rulebook
- Product Codes Listing
- Register for CME ClearPort
- Trading Hours
- View CME Clearport product slate

E-mini Crude Oil Futures

Code	QM
Venue	CME Globex, Open Outcry (New York)
Hours (All Times are New York Time/CT)	Sunday – Friday 6:00 p.m. – 5:15 p.m. (5:00 p.m. – 4:15 p.m. Chicago Time/CT) with a 45-minute break each day beginning at 5:15 p.m. (4:15 p.m. CT)
Open Outcry	Monday – Friday 9:00 AM to 2:30 PM (8:00 AM to 1:30 PM CT)
Contract Unit	500 barrels
Pricing Quotation	U.S. dollars and cents per barrel
Minimum Fluctuation	\$0.025 per barrel
Floating Price	The Floating Price for each contract month will be equal to the NYMEX Light Sweet Crude Oil Futures contract final settlement price for the corresponding contract month on the last trading day for that contract month.
Termination of Trading	Trading in the current delivery month shall cease on the business day immediately preceding to the last day of trading in the current delivery month of the NYMEX Light Sweet Crude Oil futures contract.
Listed Contracts	Current year and 5 calendar years
Position Limits	NYMEX Position Limits
Rulebook Chapter	NYMEX Rulebook Chapter 401
Settlement Type	Financial
Exchange Rule	These contracts are listed with, and subject to, the rules and regulations of NYMEX.

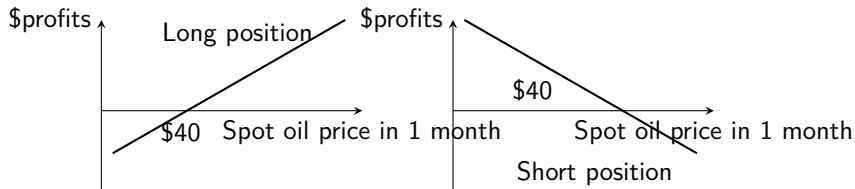
Quick Links

- Energy Update - Archive
- Excessive Speculation and Position Limits in Energy Derivatives Markets
- Free Schedule
- Free Real-time Quotes
- Learn more about CME Clearport
- Login to CME ClearPort
- Margins / Performance Bonds
- Monthly Energy Update
- NYMEX Daily Reports
- NYMEX Rulebook
- Product Codes Listing
- Register for CME ClearPort
- Trading Hours
- View CME Clearport product slate
- WTI Q2 Webinar Update: A Fundamentals Analysis
- WTI Q1 Webinar Update: Understanding the Global Crude Benchmark
- Adversity Notices
- Events Calendar
- Free Real-time Quotes
- Holiday Calendar
- Rulebooks
- Special Executive Reports
- Vendor Codes Lookup

Long and Short Positions: Profits at Maturity

Long — commitment to *purchase* the asset on the delivery date; **short** — commitment to *sell*. At maturity:

- Profit to long = spot price at maturity – original futures price
- Profit to short = original futures price – spot price at maturity



Long and short profit from opposite price movements; the futures contract is a zero-sum game — gains and losses net out to zero. **You must know how to plot profits of a derivative at maturity** (horizontal axis: spot price of the underlying at maturity).

Margin Requirements on Futures

Initial margin Deposit required on futures trades to ensure the terms of the contracts will be met

Maintenance margin Margin a trader must maintain once a position is taken — if losses push the account below it, the customer gets a **margin call** and must deposit funds to restore the *initial* margin to keep the position open

Brokers can close out a customer's position if margin is not maintained. Margin requirements are set by exchanges and vary across contracts (and over time).

Example: 30-Year Treasury Bond Futures (CBOT)

Contract	Exchange	Delivery Months	Contract Size	Deliverable Instrument	IMR	MMR
T-Bond	CBOT	M, J, S, D	\$100,000 face value	T-bonds maturing at least 15 years (and no more than 25 years) from the delivery date	\$4,070	\$3,700

Going **long** one June contract = agreeing to buy \$100,000 face value of T-bonds at the original futures price at June expiry; going **short** = agreeing to deliver the bonds and receive that price. Each investor posts the IMR of \$4,070 at initiation and must maintain the MMR of \$3,700 while the position is open.

Marking to Market: Example

IMR \$4,070, MMR \$3,700. You buy one June contract at the opening price of 98'16 ($98 + 16/32 = 98.5$); Monday settles at 98'10 (98.3125), Tuesday closes at 97'00. What is in your margin account after Tuesday's settle?

Long	Settle	Underlying Value	Price Change	Margin Account
Open	—	\$98,500.00	—	\$4,070.00
Mon.	98'10	\$98,312.50	(−\$187.50)	\$3,882.50
Tues.	97'00	\$97,000.00	(−\$1,312.50)	\$2,570.00
<i>Margin call (below \$3,700): add cash</i>				\$1,500.00
				\$4,070.00

Who receives the money taken out of your margin account? The counterparties who gain (via the clearing house).

Valuation of Futures and the Parity Condition

- The futures price (F) is always linked to the spot price (S) of the underlying asset
- The opportunity costs or benefits of owning the futures instead of the underlying asset itself causes the futures price to be different from the spot price
- Example of such benefits (+) and costs (-) of futures:
 - + Carrying costs (e.g. storage costs $c\%$)
 - + Opportunity costs of money (risk-free interest rate $r\%$)
 - – Dividend yield ($d\%$) of owning underlying asset
 - – Convenience yield ($v\%$) of owning underlying asset
- The general formula is:

$$F = S(1 + r + c - d - v)^T$$

Where r , d , c , and v are in percentage per annum. T is the maturity of the futures contract

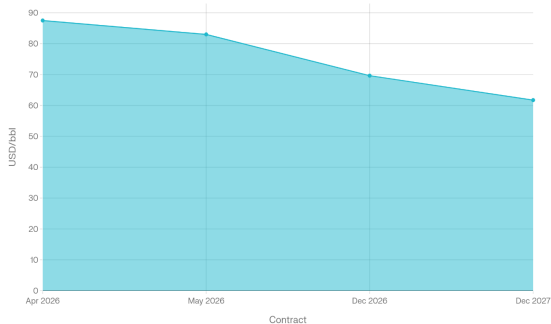
- Intuition: $F > S$ if the net benefits of owning futures are larger than owning underlying asset directly, so futures price will be higher than spot price
- E.g. A 30-day single stock futures price is affected by interest rate and dividend yield. No carrying cost or convenience yield.
 - Interest rate effect implies that futures price should be higher than the spot. By not buying the underlying asset, we don't need to tie up cash in asset, hence this cash on hand can earn interest (r). Since one is better off holding the futures, one should pay more for the futures.
 - Dividend yield effect implies that futures price should be lower than the spot price. By buying the futures, we are not entitled to any dividends (d) that are paid out by the underlying asset. Since one is worse off with the futures, one should only pay a lower price for the futures.
 - Overall, we get $F = S(1 + r - d)^T$
- Futures price > Spot price is called contango (usually true for commodities)
- Futures price < Spot price is called backwardation

Contango vs. Backwardation

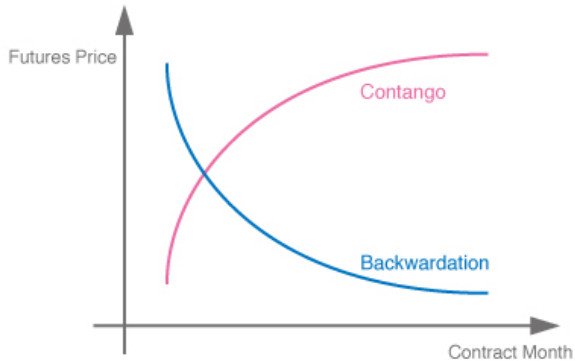
WTI futures curve (Mar 10, 2026)

Source: Barchart, Premium Feed Grain | verified maturities show backwardation

Powered by  perplexity



Relationship with Contract Month and Futures Price



Basis = spot price – futures price. The two converge as maturity approaches.

- Difference between the futures and spot price is called the “basis”.
- The futures price will converge towards the spot price at maturity because as maturity approaches, $(1 + r + c - d - v)^T$ becomes closer and closer to 1 as T goes to 0.
- E.g. suppose futures > spot (contango) one month before maturity. As maturity approaches, the futures price will converge to the spot price.

Arbitrage with Futures

Riskless profit is possible when the actual futures price differs from the price implied by the futures parity condition.

Example. Gold costs 2% p.a. to store; the interest rate is 4% p.a.; spot gold is \$4,000/oz (Sep 2026). The one-year futures price should be

$$F = 4000 \times (1 + 0.02 + 0.04) = \$4,240$$

If the actual futures price is too high, e.g., \$4,250: borrow \$4,000, buy spot gold now (store for one year), and enter a futures contract to sell gold at \$4,250 in one year:

$$\text{Profit (at maturity)} = -\$4000 \times (1 + 2\% + 4\%) + \$4250 = \$10$$

Question: what if the actual futures price is too low, e.g., \$4,220?

Risks of Trading Futures

- Market risk** Speculators win or lose on price changes; hedgers hold the underlying and are unaffected by contract price volatility — unless they cannot meet margin calls before expiration
- Basis risk** Imperfect correlation between % changes in futures and spot prices over the hedging period, producing an imperfect hedge (e.g., airlines hedging jet fuel with crude oil futures)
- Liquidity risk** Thinly traded contracts make it hard to find a counterparty to close a position before maturity
- Counterparty risk** High for forwards (default by the other party can cause large losses); mitigated in futures by daily marking to market and the clearing-house guarantee

Options: Rights, Not Obligations

Options give the buyer the **right but not the obligation** to buy or sell the underlying at a pre-specified price (the **strike price**) within a specific period. The buyer pays the seller a **premium**.

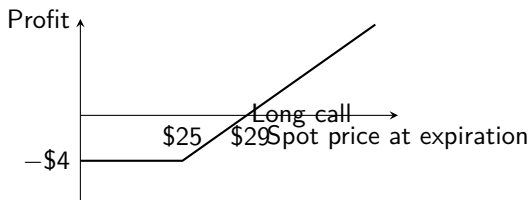
- **Call** = right to buy; **put** = right to sell
- The **writer** (seller) receives the premium upfront and is *obliged* to sell (call) or buy (put) if the buyer exercises
- Unlike futures/forwards: the option costs a premium, and the buyer can “walk away” — losing only the premium
- **European** options: exercisable only at maturity; **American**: anytime up to maturity (most equity options are American)

Moneyness: spot = strike $\rightarrow$ “at the money.” *In the money*: exercising immediately is profitable — calls when $S > X$; puts when $S < X$. *Out of the money*: the reverse.

Speculating with Call Options

Buy a call when you think the spot price will rise above strike X : pay the premium for the right, but not the obligation, to buy at X .

- $S > X$ at maturity: in the money — exercise. Net profit = spot at maturity — strike — premium
- $S \leq X$: out of the money — do not exercise; lose only the premium

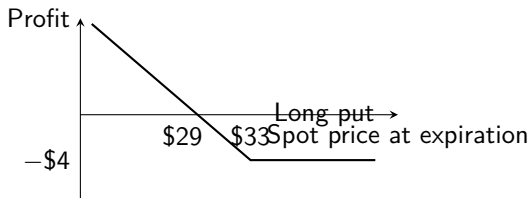


Long 1 call with strike \$25, premium \$4: loss capped at -\$4; break-even at \$29; unlimited upside.

Speculating with Put Options

Buy a put when you think the spot price will fall below strike X : pay the premium for the right, but not the obligation, to sell at X .

- $S < X$ at maturity: in the money — exercise. Net profit = strike — spot at maturity — premium
- $S \geq X$: out of the money — do not exercise; lose only the premium



Long 1 put with strike \$33, premium \$4: loss capped at -\$4; break-even at \$29; maximum gain \$29 (if the spot goes to zero).

Option Profit Profiles (All Four Positions)

At maturity, with strike price X , call premium c , put premium p :

Long call Loss limited to $-c$ for $S_T \leq X$, then +1-for-1 upside — unlimited gain potential

Short call Gain limited to $+c$ for $S_T \leq X$, then -1-for-1 — unlimited loss potential

Long put Gain up to $X - p$ (when $S_T = 0$), loss limited to $-p$ for $S_T \geq X$

Short put Loss up to $-(X - p)$ (when $S_T = 0$), gain limited to $+p$ for $S_T \geq X$

All four are kinked at the strike price; buyers' losses are limited to the premium, writers' gains are limited to the premium.

Intrinsic Value and Time Value

The option premium has an **intrinsic** and an **extrinsic** component:

- Intrinsic value: calls $\max\{0, S - X\}$; puts $\max\{0, X - S\}$ (X = strike, S = spot) — determined by moneyness
- Time value = premium — intrinsic value: the option has time value because the intrinsic value could still increase before maturity; it depends on volatility of the underlying, time to maturity, dividend yield, and interest rates



Figure 10–9: The intrinsic value versus the before-exercise value of a call option

Intrinsic and Time Value: Example

MSFT — underlying stock price \$460.00; expiry Sep 18, 2026				
Type	Strike	Last	Bid / Ask	IV
Call	460	34.84	34.30 / 35.95	38.4%
Put	495	3.68	3.60 / 3.90	23.7%

With MSFT trading at \$460, the Sep \$460-strike call is *at the money*: intrinsic value $= \max\{0, 460 - 460\} = \0 , so its entire \$34.84 premium is *time value*. Puts work the mirror way: a put is in the money only when $S < X$ — the right to sell above \$460 is what would make a put strike such as \$495 valuable. Note the bid–ask spread: buyers pay the ask (\$35.95), sellers receive the bid (\$34.30) — the spread is a round-trip trading cost.

Determinants of Option Prices

Black and Scholes (1973) derived the formula for pricing an option from observable factors. Directional effects on the *total* premium (intrinsic + extrinsic):

Factor	Call	Put
Current price of underlying	+	−
Strike price	−	+
Yield from holding underlying asset (div)	−	+
Time to maturity	+	+
Price volatility of underlying asset	+	+
Interest rate	+	−

Reasoning: longer maturity $\Rightarrow$ more chance of getting in the money (both). Higher volatility $\Rightarrow$ options profit from favorable moves while downside is limited (both). Higher interest rates make calls more valuable (defer paying the spot price; invest at the high rate) and puts less valuable (the PV of the strike falls). Expected dividends push the stock price down on the ex-date while the strike is fixed $\Rightarrow$ calls less, puts more likely in the money.

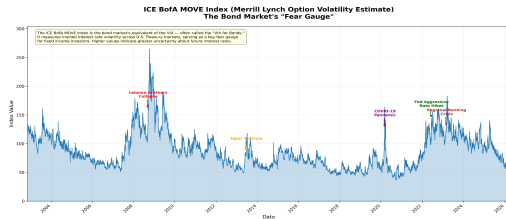
Volatility Indices: VIX and MOVE

Option prices can be used to back out investors' expectations of market volatility (**implied volatility** — forward-looking).

The VIX: real-time index of market expectations for S&P 500 volatility over the next 30 days — the “fear gauge,” rising sharply during market stress. As of early September 2026 it sits near 14.5, versus a 12-month range of 13.4–35.3 (the high came with the March 2026 Iran-conflict spike).



ICE BofA MOVE Index: the bond market's “VIX for bonds” — implied interest-rate volatility across U.S. Treasury markets; it can *lead* equity volatility, sometimes signaling stress before the VIX. In late summer 2026 it trades near 74, within a trailing 12-month range of 40.7–108.8.



Hedging: First, Are You Long or Short the Underlying?

Derivatives let investors and firms remove (hedge away) risks they do not want — with futures/forwards or with options. To hedge, **take the opposite of your underlying spot position** in a futures/forward contract. So: is your underlying position long or short?

General rule: you are **LONG** “something” (currency, stock, real estate, commodity) if you are happy when its price goes **up**; **SHORT** if you are happy when it goes **down**. Especially useful for natural positions from business or expected future actions:

- You own a car — long or short oil?
- You will buy an apartment in HK next year — long or short HK real estate?
- A farmer growing wheat — long or short wheat?

Hedging with Futures: Example

An investor must sell a stock in six months to fund a house down-payment. The stock is worth \$50 now; his profit/loss over the next six months is uncertain. **His underlying position is long the stock.**

Futures hedge: short a futures contract on the stock. Assume the futures price is exactly \$50 (theoretical futures price: $F = S(1 + r - d)$).

- If the stock rises, he gains on the underlying but loses on the short futures — gives up the upside
- If the stock falls, he loses on the underlying but gains on the short futures — sheltered from the downside
- The hedge **completely eliminates** price risk of the underlying: the hedged profit profile is a flat line at \$0 (relative to today's \$50 price)

Result of the Futures Hedge

Hedged position = underlying (long stock, profit line through \$50) + short futures (mirror image): the combined profit is a **flat line** — locked in at the futures price of \$50.

What is the point? Why not just sell the stock now?

- May be unable to sell (restrictions, or the underlying position is only expected — the house purchase is future)
- Transaction costs of selling and repurchasing can exceed the hedge
- Under the futures parity condition, the hedge is equivalent to selling — same expected payoff
- Flexibility: futures positions can be adjusted far more easily than the spot position

Basis Risk in Futures Hedging

The textbook hedge is almost too good to be true. In practice:

- The asset being hedged may differ from the asset underlying the futures contract
- The exact date of the future purchase/sale may be uncertain
- The hedge may require closing the futures position before its delivery month

These give rise to **basis risk**: the spot–futures spread may widen or narrow between initiation and liquidation of the hedge.

Example: if the investor had to sell in three months, with spot at \$45 and futures at \$47, the net amount received is $\$45 + (\$50 - \$47) = \48 — less than the expected \$50.

Hedging with Options: Retain Some Upside

Can the investor keep some upside while limiting downside? Yes — with options. Buy a 6-month **put** with strike $K = \$50$, premium \$4:

Stock at maturity	Long profit	stock	Exercise ($K=\$50$)	put?	Option profit (incl. premium)	Total
30	-20		Y		$20 - 4$	-4
40	-10		Y		$10 - 4$	-4
50	0		Indifferent		-4	-4
54	+4		N		-4	0
60	+10		N		-4	+6
70	+20		N		-4	+16

Worst case limited to $-\$4$ (the premium); upside retained beyond \$54.

Hedging a Long Position with a Long Put

Combined profile = (1) underlying long stock + (2) long put = (3) options hedge:

- For stock prices at maturity below \$50: the put pays off — total profit flat at $-\$4$ (the put premium)
- For stock prices above \$50: the put expires worthless — total profit rises one-for-one with the stock, shifted down by \$4, breaking even at \$54
- Compared with the futures hedge (a flat line at \$50), the options hedge **retains the upside** at the cost of the \$4 premium

Using Options to Hedge the Underlying

Long asset, long put Hedge a *long* position with a long put (the previous example): downside limited at $-p$, upside retained

Short asset, long call Hedge a *short* position with a long call: losses on the short asset above K are offset by the call; downside of the hedge limited at $-c$

Assume option strike = asset price = K , put premium = p , call premium = c . The two combined profiles are mirror images: each floors the worst case at the premium paid while leaving one side of the price movement open.

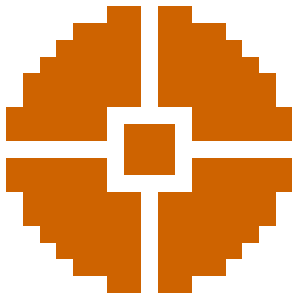
Speculating with Options

Options are particularly suitable for speculators betting on *sophisticated* views of future price movement. With futures/forwards you can only bet on a rise or a fall; with options you can bet on:

- A **large** rise/drop in the price of the underlying security
- A **moderate** rise/drop in the price of the underlying security
- The price moving **a lot in either direction** (long volatility)
- The price changing **little in either direction** (short volatility)

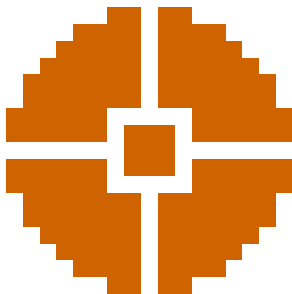
Bull Spread Using Calls

Buy a call with strike price K_1 and sell a call with strike price $K_2 > K_1$: profit is capped above K_2 and floored at the net premium. **Belief:** the underlying's price will rise, but only moderately.



Bear Spread Using Puts

Buy a put with strike price K_2 and sell a put with strike price $K_1 < K_2$: profit is capped below K_1 . **Belief:** the underlying's price will fall, but only moderately.



Collar and Straddle

Collar An options strategy that brackets the value of an asset between two bounds: long the underlying, buy an out-of-the-money **put** to limit downside, fund it by writing an out-of-the-money **call** (also limiting upside). Net outlay for the options is approximately zero. (Madoff claimed to use this strategy.)

Straddle Long call + long put at the *same* strike and maturity — a bet on volatility. To profit, the stock price must move by more than the cost of both options, in *either* direction. The writer of a straddle bets the price will not move much. Nick Leeson's short straddle caused the collapse of Barings Bank in January 1995; a modern parallel is Archegos (2021), whose leveraged total-return-swap positions imploded for \$36 billion of bank losses — founder Sung Kook (Bill) Hwang was sentenced to 18 years in November 2024.

Put-Call Parity

The call-plus-bond portfolio must cost the same as the stock-plus-put portfolio:

$$C + \frac{X}{(1+r)^T} = S + P$$

where C and P price a call and put with the same maturity T and strike X . At maturity both portfolios yield the same payoff regardless of S_T :

- If $S_T > X$: the call is exercised (left $\rightarrow$ stock); the put expires worthless (right $\rightarrow$ stock)
- If $S_T < X$: the call expires worthless (left $\rightarrow$ receive X from the bond); the put is exercised, selling the stock at X (right $\rightarrow X$)

$$C + \frac{X}{(1+r_f)^T} = S_0 + P$$

Violation of Put-Call Parity: Example

$S = 110$, $C = 17$, $P = 5$, risk-free rate 4%, maturity 1 year, $X = 104$:

$$C + X/(1 + r) = 17 + 100 > S + P = 110 + 5$$

The right side is cheaper: buy the stock and a put, sell the call, and borrow the PV of the strike ($\$104/1.04 = \100):

Position	Immediate CF	CF in 1 yr, $S_T < 104$	CF in 1 yr, $S_T \geq 104$
Buy stock	-110	S_T	S_T
Borrow $\$104/1.04 = \100	+100	-104	-104
Sell a call option	+17	0	$-(S_T - 104)$
Buy a put option	-5	$104 - S_T$	0
Total	+2	0	0

Riskless profit of \$2 regardless of the stock price at maturity.

Key Takeaways: Derivatives and Futures

Derivatives Securities with value derived from an underlying asset (stock, currency, commodity); used for speculation, hedging, or arbitrage; zero-sum game

Forwards vs. futures Forwards: customized, OTC, counterparty risk, settled at maturity.
Futures: standardized, exchange-traded, marked to market daily, clearing-house guarantee, margin requirements

Futures payoffs Long: $\text{profit} = \text{spot} - \text{futures price}$ (unlimited upside); short: $\text{futures price} - \text{spot}$ (unlimited downside)

Key Takeaways: Options and Hedging

Options Rights, not obligations. Call: right to buy; put: right to sell. Buyer pays premium — limited loss, unlimited gain; writer receives premium — limited gain, unlimited loss. Premium driven by underlying price, strike, maturity, volatility, interest rates

Hedging Futures: take the opposite position to the underlying exposure — eliminates risk and upside. Options: long put protects a long asset; long call protects a short asset

Parity conditions Put-call: $C + X/(1 + r)^T = S + P$; futures: $F = S(1 + r)$, with dividend yield $F = S(1 + r - d)$